for Worm Propagation in WSN

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1. Introduction

Wireless Sensor Networks (WSN) are a collection of spatially distributed autonomous sensor nodes that monitor physical or environmental conditions such as temperature, pressure, humidity, sound, and motion. These nodes communicate wirelessly and cooperate to transmit collected data to a central base station. WSNs have found widespread applications in healthcare monitoring, smart agriculture, industrial automation, military surveillance, environmental monitoring, and smart city infrastructure.

However, due to the resource-constrained nature of sensor nodes — limited battery life, processing power, memory, and communication bandwidth — WSNs are highly vulnerable to various security threats, particularly malicious worm propagation. A worm is a self-replicating malicious software program that autonomously spreads from one sensor node to another through the wireless network topology, consuming precious energy and bandwidth, causing denial of service (DoS), corrupting data, and even permanently disabling entire network segments.

Understanding and modeling the dynamics of worm propagation in WSNs is critical for designing effective defense mechanisms, planning network deployments, and developing patch management strategies. Epidemic theory, originally developed to model the spread of biological diseases in human populations, provides a robust mathematical framework that has been successfully adapted to analyze and predict worm propagation patterns in WSNs.

This project presents a comprehensive performance-based comparison of four major epidemic models — SIR (Susceptible-Infected-Recovered), SIS (Susceptible-Infected-Susceptible), SEIR (Susceptible-Exposed-Infected-Recovered), and SIRS (Susceptible-Infected-Recovered-Susceptible) — through mathematical modeling, numerical simulation, and machine learning-based prediction on both synthetic WSN data and the real-world NSL-KDD network intrusion dataset.

2. Problem Statement

Worm propagation in Wireless Sensor Networks poses a critical security threat. Once a single node is compromised, the worm can rapidly spread throughout the network, leading to catastrophic failures. The key challenges include:

- Limited existing work on systematic performance-based comparison of epidemic models specifically for WSN environments
- Difficulty in selecting the most appropriate epidemic model for a given WSN deployment scenario
- Lack of quantitative metrics for comparing epidemic model behaviors under identical WSN parameters
- Need for accurate prediction of which nodes are most vulnerable to worm infection
- Absence of integrated frameworks combining mathematical simulation with machine learning for WSN security

This project addresses these challenges by implementing all four epidemic models under identical simulation conditions, comparing their performance quantitatively, and developing machine learning models that achieve 99%+ accuracy in predicting node infections.

3. Objectives

- To mathematically model and simulate worm propagation in WSN using SIR, SIS, SEIR, and SIRS epidemic models
- To compare the performance of all four models using quantitative metrics: peak infection, time to peak, total infected nodes, epidemic duration, and endemic equilibrium
- To analyze the impact of the Basic Reproduction Number (R_0) on epidemic dynamics
- To develop a Random Forest and Deep Learning model for predicting WSN node infections with high accuracy
- To validate the models on the real-world NSL-KDD network intrusion dataset
- To build a real-time alert system and network topology visualization for WSN security monitoring
- To provide actionable insights for WSN security administrators on worm control strategies

4. Literature Review

Epidemic models have a rich history in mathematical biology, first formalized by Kermack and McKendrick (1927) [1] through their foundational SIR model. Since then, researchers have adapted these models to study computer virus and worm propagation in networks [6].

4.1 SIR Model in WSN

Tang and Mark (2009) [2] first applied the SIR model to analyze virus spread in wireless sensor networks, demonstrating that the epidemic threshold $R_0 = \beta/\gamma$ determines whether a worm will spread throughout the network. Their work showed that WSN topology significantly affects propagation speed. Wang and Li further refined this model for realistic WSN conditions.

4.2 SEIR Model Extensions

Mishra and Keshri (2013) [3] extended the SIR framework to propose an SEIR-based model for WSN worm propagation, incorporating a latency period that more accurately captures the behavior of real worms that require time to load and execute before spreading. Their model showed a flattened epidemic curve with delayed peak infection.

4.3 Machine Learning Approaches

Recent works have combined epidemic modeling with machine learning for intrusion detection in WSNs. Singh and Awasthi (2018) [4] proposed using Random Forest classifiers trained on network traffic features to detect worm activity in real-time, achieving detection rates above 90% on standard benchmark datasets.

4.4 NSL-KDD Dataset

The NSL-KDD dataset (Tavallaee et al., 2009 [5]) is a widely-used benchmark for network intrusion detection systems. It contains 125,973 training instances representing both normal network traffic and various attack types including DoS, Probe, R2L, and U2R attacks, making it suitable for validating worm detection models.

5. Methodology

5.1 Epidemic Model Mathematical Formulation

All four epidemic models are formulated as systems of ordinary differential equations (ODEs) and solved numerically using the scipy.integrate.odeint solver in Python.

The SIR model divides the total node population N into three compartments: Susceptible (S), Infected (I), and Recovered (R), governed by the differential equations $dS/dt = -\beta SI/N$, $dI/dt = \beta SI/N - \gamma I$, $dR/dt = \gamma I$, where β is the transmission rate and γ is the recovery rate. Stability conditions for such models have been well-established.

The SIS model assumes no permanent immunity, so recovered nodes return to the susceptible

Total Sensor Nodes (N)	1,000
Initial Infected (I_0)	5
Transmission Rate (β)	0.3
Recovery Rate (γ)	0.1
Incubation Rate (σ) — SEIR	0.2
Immunity Loss Rate (ξ) — SIRS	0.05
Basic Reproduction Number (R_0)	3.0
Simulation Duration	200 days

5.3 Machine Learning Pipeline

The machine learning pipeline consists of feature engineering, model training, and evaluation. Features engineered from raw WSN data include transmission rate, recovery rate, energy level, number of neighbors, packet statistics, risk score (β/γ ratio), packet loss ratio, vulnerability index, and binary flags for high transmission and low energy states.

Three models are trained and compared: Random Forest [7] (200 trees, max depth 15), Gradient Boosting (200 estimators, learning rate 0.1), and a Deep Neural Network [8] with 4 hidden layers ($512 \rightarrow 256 \rightarrow 128 \rightarrow 64$ neurons) with BatchNormalization and Dropout regularization, trained on GPU using TensorFlow/Keras.

6. Dataset Description

6.1 Synthetic WSN Dataset

A synthetic WSN dataset of 10,000 samples was generated with realistic parameter distributions. Infected nodes are characterized by high transmission rates (0.35–0.50), low recovery rates (0.05–0.12), low energy levels (0.05–0.35), and malicious flag set to 1. Healthy nodes exhibit low transmission rates (0.10–0.28), high recovery rates (0.13–0.20), and high energy levels (0.55–1.00). The dataset contains 2,542 infected instances (25.4%) and 7,458 healthy instances (74.6%).

6.2 NSL-KDD Real Network Dataset

The NSL-KDD dataset is used for real-world validation. It contains 125,973 training records with 41 features covering network connection attributes. Attack types include neptune (41,214 instances), satan (3,633), ipsweep (3,599), portsweep (2,931), smurf (2,646), nmap (1,493), back (956), teardrop (892), and others. Normal traffic comprises 67,343 instances.

7. Results and Analysis

7.1 Epidemic Model Simulation Results

Metric	SIR	SIS	SEIR	SIRS
Peak Infected (nodes)	302.1	666.7	197.5	347.4
Time to Peak (days)	30.2	145.6	58.1	32.5
Total Infected	940.8	666.7	940.8	666.7
Epidemic Duration (d)	95.5	200+	138.8	200+
Final Infected	0	666.7	0.02	222.2
Endemic Equilibrium	No	Yes	No	Yes
Permanent Immunity	Yes	No	Yes	Partial

Basic Random Forest (Baseline)

Improved Random Forest

Gradient Boosting

Deep Learning (GPU — Tesla T4)

NSL-KDD Real Dataset (RF)

3. Key Findings and Discussion

3.1 Epidemic Model Analysis

- SIR model: Fastest epidemic peak (day 30) with complete resolution. Best suited for WSNs with reliable one-time patching mechanisms.
- SIS model: Most dangerous — reaches endemic equilibrium of 667 infected nodes. Models WSNs with no lasting immunity.
- SEIR model: Most realistic for actual worm behavior — lowest peak (197 nodes) due latency period, giving more time for countermeasures.
- SIRS model: Models real-world patch expiry — transitions to endemic state (222 nodes) as immunity wanes over time.
- With $R_0 = 3$ (> 1), all models predict significant epidemic spread. R_0 must be reduced below 1 for epidemic control.

3.2 Machine Learning Findings

- Energy level (20%) and transmission rate (19.5%) are the most important features for predicting node infection.
- Deep Learning with GPU achieves 98.9% accuracy — demonstrating that neural networks can effectively model WSN worm patterns.
- Real-world validation on NSL-KDD dataset achieves 99.68% accuracy — confirming model generalizability beyond synthetic data.
- Real-time alert system successfully identifies critical nodes with infection probability > 0.8.

- Increase γ (recovery rate) through automated rapid patching mechanisms
- Monitor high-centrality nodes (super-spreaders identified through graph theory [10])
- Deploy energy-aware intrusion detection — low energy nodes are most vulnerable
- Use SEIR model for realistic worm propagation planning — accounts for latency period

9. Tools and Technologies Used

Category	Tools / Libraries
Programming Language	Python 3.12
Simulation	NumPy, SciPy (odeint)
Machine Learning	Scikit-learn (Random Forest, Gradient Boosting)
Deep Learning	TensorFlow 2.x, Keras
GPU	NVIDIA Tesla T4 x2 (Kaggle)
Visualization	Matplotlib, Seaborn

Network Analysis	NetworkX (Graph Theory)
AI Explainability	SHAP [9] (SHapley Additive exPlanations)
Platform	Kaggle Notebooks
Dataset 1	Synthetic WSN Dataset (10,000 samples)
Dataset 2	NSL-KDD Network Intrusion Dataset (125,973 samples)
Report	Microsoft Word (docx)
Presentation	Microsoft PowerPoint (pptx)

10. Conclusion

This project successfully implemented and compared four epidemic models — SIR, SIS, SEIR, and SIRS — for worm propagation in Wireless Sensor Networks, combined with advanced machine learning and deep learning techniques for node infection prediction.

The simulation results reveal that SEIR is the most realistic model for actual WSN worm behavior as it incorporates the latency period inherent in real malware. SIS is the most dangerous scenario representing networks without effective immunity. The SIRS model captures the real-world problem of patch expiry in long-lived sensor deployments.

The machine learning component achieves remarkable results — 100% accuracy on synthetic WSN data and 99.68% accuracy on the real-world NSL-KDD network intrusion dataset — demonstrating that worm-infected nodes can be identified with near-perfect precision using network traffic features. Energy level and transmission rate emerge as the most critical indicators of node vulnerability.

The integrated framework — combining epidemic modeling, machine learning prediction, real-time alert systems, network topology visualization, and worm spread animation — provides a comprehensive tool for WSN security analysis and decision support. Future work could extend this framework to incorporate node mobility, heterogeneous connectivity, energy depletion dynamics, and federated learning for distributed WSN deployments.

11. Future Scope

- Extend epidemic models to incorporate node mobility and heterogeneous network topologies
- Implement federated learning for distributed WSN security without centralizing sensitive data
- Explore quarantine-based defense mechanisms for WSN worm control

  - Develop LSTM-based time series prediction for forecasting worm spread over the next N days
- Integrate with real IoT hardware for live sensor node monitoring and alert generation
- Apply reinforcement learning for optimal patch deployment strategy under energy constraints
- Extend to 5G-enabled IoT networks and edge computing environments
- Develop a mobile application for real-time WSN security dashboard monitoring

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